International Journal of Number Theory

A Weighted Turán Sieve for Twin Primes via the Krafft Geometry

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Received (Day Month Year)
Revised (Day Month Year)
Accepted (Day Month Year)
Published (Day Month Year)

The Twin Prime Conjecture is obstructed by the parity barrier, which prevents classical multiplicative sieves from isolating prime pairs. We introduce an additive Turán variance framework over bounded, symmetric intervals. Using the discrete Krafft alignment, we construct an isomorphism under which a local penalty indicator governs twin prime existence. Passing to the frequency domain reveals a negative resonance at the third harmonic. We reduce the Twin Prime Conjecture to a finite-dimensional inequality: the bound $\mu_{\min}(n) < 1$ implies the existence of twin primes in a prescribed interval. We further show that independent 1-dimensional weights cannot overcome the logarithmic divergence of Mertens' sums, forcing the quotient to $\mu \geq 1$. This density deficit is a spectral manifestation of the parity barrier, indicating that a proof requires multidimensional cross-harmonic polynomials.

Keywords: Twin Prime Conjecture, Turán sieve, Krafft geometry, Parity barrier, Discrete Fourier Transform

Mathematics Subject Classification 2020: 11N05, 11N35, 11N36

1. Introduction

The Twin Prime Conjecture, which asserts the existence of infinitely many prime pairs differing by two, is one of the central open problems in number theory. Classical benchmarks for its density were established by the Hardy–Littlewood conjecture [8]. Sieve methods, originating with Brun [2] and refined by Selberg [13] and others [11], have been used to approach questions of prime distribution. While these multiplicative inclusion-exclusion frameworks have yielded significant structural results—most notably strictly bounding the gaps between primes [3, 4, 6, 10]—they encounter the parity barrier, a combinatorial obstruction that prevents classical sieves from distinguishing integers with an even number of prime factors from those with an odd number.

To bypass this barrier, this paper moves from a multiplicative inclusion-exclusion model toward an additive Turán variance framework [15]. Rather than counting survivors destructively via a product, we constrain our search to highly structured symmetric intervals $\mathcal{A}_n = [6n^2 - 2n, 6n^2 + 10n + 3]$. Following the discrete Krafft geometry [9, 12], our framework utilizes the alignment residue $r_i^K = \lfloor (p_i + 1)/6 \rfloor$ for primes in our bounded sieve window. Our additive sieve isomorphism (formally verified in Lean) shows that any surviving integer corresponds to a twin prime pair.

Translating this discrete spatial model into the continuous frequency domain via the Discrete Fourier Transform and Parseval's identity reveals that the previously specified Krafft alignment forces a universal negative resonance precisely at the third harmonic. We perform a topological reduction over the space of polynomial weights. We evaluate a minimum spectral ratio $\mu_{\min}(n)$, formally reducing the Twin Prime problem to a strictly finite- dimensional conditional inequality: assuming the analytic bound $\mu_{\min}(n) < 1$ holds, twin primes exist within $\mathcal{A}_n$.

Furthermore, we also provide an explicit analytic result regarding the limitations of 1-dimensional harmonic weights. We prove that deploying purely independent, 1-dimensional harmonic weights inherently fails to overcome the local density bounds—specifically, because bounded independent phase corrections cannot structurally outpace the logarithmic divergence of Mertens' sums. This forces the quotient to $\mu \geq 1$. This density deficit is a spectral form of the parity obstruction, indicating that resolving the conjecture relies intrinsically on the full $2^{w(n)}$ dimensionality of polynomial cross-harmonics to produce a favorable sieve density.

Throughout the text, ∀ symbols are used as direct links to formalized code in the GitHub repository `https://github.com/ElNando888/KrafftSieve/tree/v1.2`, which contains the machine-checked verifications for all definitions, statements, and proofs in this paper.[a] We use Lean [5] version 4.28.0 together with the Lean mathematical library `Mathlib` [14] revision `8f9d9cf` (dated 2026-02-16).

2. Discrete Geometry and the Additive Sieve Isomorphism

To circumvent the parity limitations inherent in multiplicative sieves, we transition to an additive framework evaluated over a strictly bounded, symmetric interval. We begin by formalizing the specific geometry of our prime window and evaluation space.

Definition 2.1 (The Prime Window and Primorial [∀ & ∀]). Let $\mathcal{P}_n$ denote the set of primes evaluated by the sieve, bounded strictly as:

$$\mathcal{P}_n = \{p \text{ prime} \mid 5 \leq p < 6n + 2\}$$

We define $w(n) = |\mathcal{P}_n|$ as the total count of these prime factors, and $q = \prod_{p \in \mathcal{P}_n} p$ as their associated primorial.

Definition 2.2 (The Krafft Residues [∀]). For each prime $p_i \in \mathcal{P}_n$, we establish

[a]The only axioms used are `propext`, `Classical.choice`, and `Quot.sound`, which can be checked by the `#print axioms` command.

the target Krafft alignment residue as:

$$r_i^K = \left\lfloor \frac{p_i + 1}{6} \right\rfloor$$

Definition 2.3 (The Evaluation Interval [ ]). We constrain our search for twin primes to the highly structured symmetric interval $\mathcal{A}_n$:

$$\mathcal{A}_n = [6n^2 - 2n, 6n^2 + 10n + 3]$$

This specific interval width is chosen so that the Krafft alignment residue r_i^K effectively translates the standard divisibility conditions of twin prime candidates into a single modular congruence evaluated at the central coordinate x.

Lemma 2.4 (Krafft Algebraic Equivalence [ ]). *For any prime factor $p \geq 5$ and any integer x, let the Krafft alignment residue be $r = \lfloor (p+1)/6 \rfloor$. Then $x \equiv \pm r \pmod{p}$ if and only if $p \mid (6x - 1)$ or $p \mid (6x + 1)$.*

Proof. Because $p \geq 5$, it must satisfy $p \equiv 1 \pmod{6}$ or $p \equiv 5 \pmod{6}$. Consequently, evaluating the scale modulo p yields $6r \equiv 1 \pmod{p}$ or $6r \equiv -1 \pmod{p}$, respectively. Multiplying the target coordinate congruence $x \equiv \pm r \pmod{p}$ by 6 gives $6x \equiv \pm 6r \equiv \pm 1 \pmod{p}$, rigorously verifying the divisibility rules. □

To track the sieve's progress additively, we establish a global penalty that accumulates across all primes in our window.

Definition 2.5 (Local and Global Hits [&]). We characterize local hits through the indicator function $g_i(x) \in \{0, 1\}$. For an integer x, a hit occurs if it falls into the forbidden Krafft residue classes modulo p_i:

$$g_i(x) = \begin{cases} 1 & \text{if } x \equiv \pm r_i^K \pmod{p_i} \\ 0 & \text{otherwise} \end{cases}$$

The global sieve penalty $c(x)$ is defined additively as the sum of all local hits across the primorial base:

$$c(x) = \sum_{i=1}^{w(n)} g_i(x)$$

Unlike classical inclusion-exclusion setups, $c(x)$ is non-negative; an integer with zero hits has successfully avoided all forbidden congruences. To ensure that surviving this localized sieve guarantees true primality, we rely on the strict algebraic boundaries of $\mathcal{A}_n$.

Lemma 2.6 (Interval Projection Bound [ ]). *For any integer coordinate $x \in \mathcal{A}_n$, the projected upper bound $6x + 1$ is strictly less than the square of the next possible prime candidate after the sieve window: $6x + 1 < (6n + 5)^2$.*

Proof. The maximum possible coordinate in $\mathcal{A}_n$ is $x = 6n^2 + 10n + 3$. Evaluating the projection yields $6(6n^2 + 10n + 3) + 1 = 36n^2 + 60n + 19$. Comparing this to the squared bound $(6n + 5)^2 = 36n^2 + 60n + 25$, the strict inequality holds trivially. $\square$

This structural limit ensures that survivors are prime, which underpins the reduction.

Theorem 2.7 (Additive Sieve Isomorphism [[illegible]]). *For any integer $n \geq 1$ and $x \in \mathcal{A}_n$, the global hit count evaluates to exactly zero, $c(x) = 0$, if and only if both $6x - 1$ and $6x + 1$ are prime numbers.*

Proof. By the definition of the global hit counter, $c(x) = 0$ implies no prime $p_i \in \mathcal{P}_n$ generates a local hit, meaning $x \not\equiv \pm r_i^K \pmod{p_i}$. By the Krafft Algebraic Equivalence, this guarantees that neither $6x - 1$ nor $6x + 1$ has any prime divisors in $\mathcal{P}_n$. Since integers of the form $6x \pm 1$ are intrinsically coprime to 2 and 3, they have no prime factors strictly less than $6n + 5$. Because Lemma 2.6 structurally guarantees that the maximum possible value of $6x + 1$ is strictly less than $(6n + 5)^2$, lacking prime factors up to this threshold unconditionally forces primality. $\square$

3. Fourier Expansion and the Resonant Sieve Equation

To analyze the distribution of twin prime indices $x \in \mathcal{A}_n$ that survive the local hit penalties, we pass to the frequency domain. We introduce an arbitrary, real-valued weight function $W(x)$ strictly supported within our bounded interval.

Definition 3.1 (Weighted Masses). For an arbitrary real-valued weight function $W(x)$ supported strictly on $\mathcal{A}_n$, we track the total mass S_1 and the penalized mass S_2:

$$S_1(n, W) = \sum_{x \in \mathcal{A}_n} W(x)$$

$$S_2(n, W) = \sum_{x \in \mathcal{A}_n} W(x)c(x)$$

Lemma 3.2 (The Weighted Existence Principle [[illegible]]). *If there exists a non-negative weight function $W(x)$ such that the global penalized mass is strictly less than the base spatial mass, $S_2(n, W) < S_1(n, W)$, then there must exist at least one integer $x \in \mathcal{A}_n$ such that $W(x) > 0$ and $c(x) = 0$.*

Proof. The global hit counter $c(x)$ is a sum of non-negative indicator functions, forcing $c(x) \geq 0$. Consequently, any non-zero hit count must satisfy $c(x) \geq 1$. If $c(x) \geq 1$ for all coordinates where $W(x) > 0$, then the sum $S_2(n, W) = \sum W(x)c(x)$ must be greater than or equal to $S_1(n, W) = \sum W(x)$. This contradicts the strict inequality $S_2 < S_1$, guaranteeing the existence of a coordinate with exactly zero hits carrying strictly positive weight. $\square$

Definition 3.3 (Krafft Admissibility). The Krafft Admissibility condition holds for a given n if there exists a weight function $W : \mathbb{Z}/q\mathbb{Z} \to \mathbb{R}$ that is non-negative everywhere, strictly supported on $\mathcal{A}_n$, and satisfies the strict inequality $S_2(n, W) < S_1(n, W)$.

Theorem 3.4 (Krafft Sieve Guarantee [[illegible]]). *For any integer $n \geq 1$, if Definition 3.3 is satisfied, then there exists an integer $x \in \mathcal{A}_n$ such that $6x - 1$ and $6x + 1$ form a twin prime pair.*

Proof. If Definition 3.3 is satisfied, then Theorem 3.2 guarantees the presence of at least one target coordinate $x \in \mathcal{A}_n$ experiencing exactly zero hits ($c(x) = 0$). Under Theorem 2.7, the absence of hits strictly dictates that both projected integers, $6x-1$ and $6x + 1$, are prime. □

Definition 3.5 (Discrete Fourier Transform). We shift into the continuous frequency domain by taking the Discrete Fourier Transform over the cyclic group $\mathbb{Z}/q\mathbb{Z}$. For any function $f(x)$, its frequency coefficients are given by:

$$\hat{f}(h) = \frac{1}{q} \sum_{x=0}^{q-1} f(x) e^{-2\pi i h x/q}$$

By exploiting the strict bounds of $\mathcal{A}_n$, we can extend our spatial sums over the entire cyclic group without altering their values. This allows us to leverage Plancherel's theorem to expand the penalized mass into a Fourier series.

Lemma 3.6 (Compact Support Equivalence [[illegible]]). *Assume the weight function $W(x)$ is strictly supported within the interval $\mathcal{A}_n$, meaning $W(x) = 0$ for all $x \notin \mathcal{A}_n$. Then the total spatial mass $S_1(n, W)$ is strictly proportional to the zero-frequency Fourier coefficient of the weight function:*

$$S_1(n, W) = q\hat{W}(0)$$

Proof. Since $W(x)$ is zero outside $\mathcal{A}_n$, the sum over the interval is equivalent to the sum over the entire cyclic group $\mathbb{Z}/q\mathbb{Z}$. Evaluating Definition 3.5 at $h = 0$ yields $\hat{W}(0) = \frac{1}{q} \sum_{x \in \mathbb{Z}/q\mathbb{Z}} W(x) e^0 = \frac{1}{q} S_1(n, W)$. Multiplying by q completes the proof. □

Lemma 3.7 (Plancherel's Theorem over $\mathbb{Z}/q\mathbb{Z}$ [[illegible]]). *Let $f, g : \mathbb{Z}/q\mathbb{Z} \to \mathbb{C}$ be arbitrary complex-valued functions. The inner product in the spatial domain equals the scaled inner product in the frequency domain:*

$$\sum_{x \in \mathbb{Z}/q\mathbb{Z}} f(x)\overline{g(x)} = q \sum_{h \in \mathbb{Z}/q\mathbb{Z}} \hat{f}(h)\overline{\hat{g}(h)}$$

Proof. Expanding the Fourier transforms on the right-hand side and interchanging the order of summation via Fubini's theorem isolates the frequency variable h. Applying the standard orthogonality relation for roots of unity collapses the double sum to the diagonal $x = y$, recovering the original spatial inner product. □

Lemma 3.8 (The Plancherel Hit Expansion []). *For a weight function $W(x)$ supported strictly on $\mathcal{A}_n$, the weighted hit count $S_2(n, W)$ decomposes into the Fourier domain as:*

$$S_2(n, W) = q \sum_{i=1}^{w(n)} \sum_{h=0}^{q-1} \hat{W}(h)\overline{\hat{g}_i(h)}$$

Proof. Substitute the additive definition $c(x) = \sum g_i(x)$ into $S_2(n, W)$. Because $W(x)$ has compact support on $\mathcal{A}_n$, the sum extends to $\mathbb{Z}/q\mathbb{Z}$ without altering the value. Applying Lemma 3.7 to $W(x)$ and $c(x)$, and utilizing the linearity of the Fourier transform over the finite sum of local hits, yields the expansion. □

Because the local hit function $g_i(x)$ depends strictly on the residue of x modulo the local prime p_i, projecting it onto the full cyclic group $\mathbb{Z}/q\mathbb{Z}$ causes its Fourier transform to take the form of a Dirac comb.

Lemma 3.9 (Dirac Comb Structure [&]). *The Discrete Fourier Transform (as defined in Definition 3.5) of the local hit function, $\hat{g}_i(h)$, manifests a strict Dirac comb structure. It evaluates to zero for all frequencies except those that are precise integer multiples of q/p_i. Specifically, let $h \in \mathbb{Z}/q\mathbb{Z}$:*

(1) If $h = k(q/p_i)$ for some integer $0 \le k < p_i$, then

$$\hat{g}_i\left(k\frac{q}{p_i}\right) = \frac{2}{p_i}\cos\left(\frac{2\pi k r_i^K}{p_i}\right)$$

(2) If h is not a multiple of q/p_i, then $\hat{g}_i(h) = 0$.

Proof. The local hit function $g_i(x)$ is periodic with period p_i. When projecting this p_i-periodic function onto the full cyclic group $\mathbb{Z}/q\mathbb{Z}$, the sum over x can be factored into a sum over the residue classes modulo p_i and a geometric sum over the q/p_i repetitions. If h is not a multiple of q/p_i, the geometric sum corresponds to a non-trivial root of unity and vanishes identically. If $h = k(q/p_i)$, the geometric sum constructively interferes, contributing a factor of q/p_i. The remaining sum over the two surviving residues $x \equiv \pm r_i^K \pmod{p_i}$ yields the symmetric exponential terms $e^{i\theta} + e^{-i\theta}$, which collapse into the $2\cos(\theta)$ amplitude. □

This sparsity allows us to explicitly separate the primary density term from the targeted oscillatory interference, yielding the main identity of the sieve. When evaluating the precise sub-comb frequency at the third harmonic $h = 3q/p_i$, the trigonometric phase perfectly anchors to the Krafft alignment integer.

Lemma 3.10 (The Exact Krafft Cosine []). *For any prime factor $p_i \ge 5$, substituting the Krafft alignment residue $r_i^K = \lfloor (p_i + 1)/6 \rfloor$ evaluated precisely at the third harmonic yields a strictly negative term:*

$$\cos\left(\frac{2\pi \cdot 3 \cdot r_i^K}{p_i}\right) = -\cos\left(\frac{\pi}{p_i}\right)$$

Proof. Resolving the floor function $r_i^K = \lfloor (p_i+1)/6 \rfloor$ dictates that $r_i^K = \frac{p_i \pm 1}{6}$, uniquely determined by the parity of $p_i \pmod 6$. Evaluating the phase angle generates $\frac{6\pi}{p_i}\left(\frac{p_i \pm 1}{6}\right) = \pi \pm \frac{\pi}{p_i}$. Applying the standard trigonometric identity $\cos(\pi \pm \theta) = -\cos(\theta)$ directly yields the main term phase amplitude. □

Theorem 3.11 (The Resonant Sieve Equation [??]). *For any strictly supported weight function $W(x)$, the penalized mass isolates into a primary density term and an oscillatory interference sum:*

$$S_2(n, W) = S_1(n, W) \sum_{i=1}^{w(n)} \frac{2}{p_i} + q \sum_{i=1}^{w(n)} \sum_{k=1}^{p_i - 1} \hat{W}\left(k \frac{q}{p_i}\right) \frac{2}{p_i} \cos\left(\frac{2\pi k r_i^K}{p_i}\right) \tag{3.1}$$

Proof. Starting from Lemma 3.8, we partition the inner sum over frequencies h into the zero frequency $h = 0$ and the non-zero frequencies. Applying Lemma 3.9, all non-zero frequencies h where $\hat{g}_i(h) = 0$ are discarded. The sum collapses entirely to the sub-comb frequencies $h = k(q/p_i)$. Finally, applying Lemma 3.6 replaces $q\hat{W}(0)$ with the spatial mass $S_1(n, W)$, establishing the exact equation. □

This negative resonance provides a mechanism to suppress $S_2(n, W)$ below the base population density $S_1(n, W)$.

Lemma 3.12 (Third Harmonic Extraction [??]). *For a weight function $W(x)$ whose spectral density is restricted strictly to the base harmonic $h = 0$ and the precise third harmonics $h = 3q/p_i$, the* (3.1) *equation simplifies exactly to:*

$$S_2(n, W) = S_1(n, W) \sum_{i=1}^{w(n)} \frac{2}{p_i} - q \sum_{i=1}^{w(n)} \hat{W}\left(3 \frac{q}{p_i}\right) \frac{2}{p_i} \cos\left(\frac{\pi}{p_i}\right)$$

Proof. By the spectral restriction of W, all terms in equation (3.1) where $k \neq 3$ vanish identically. For the surviving $k = 3$ terms, applying Lemma 3.10 universally inverts the sign of the interference sum, replacing the general cosine term with $-\cos(\pi/p_i)$. □

4. The Analytic Parity Collision (1-Dimensional Bounds)

Definition 4.1 (The Sieve Density Quotient). To evaluate the structural efficiency of a given weight function, we define the continuous sieve density (or Rayleigh quotient) $\mu(n, W)$ as the ratio of the penalized mass to the total spatial mass:

$$\mu(n, W) = \frac{S_2(n, W)}{S_1(n, W)} \tag{4.1}$$

By Theorem 3.2, if we can synthesize a weight function such that $\mu(n, W) < 1$, the strict Definition 3.3 condition is satisfied.

Having isolated the universal negative Krafft resonance at the third harmonic, a natural first attempt is to use 1-dimensional harmonic weights. We can evaluate the efficacy of this approach by examining the isolated spectral capacity of the sieve.

Definition 4.2 (Main Density and Resonance Capacity [&]). For a 1-dimensional harmonic weight system evaluated over $\mathcal{P}_n$, the primary un-weighted spatial density is governed by the harmonic sum over the prime window:

$$\text{Main Term}(n) = \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i}$$

The corresponding maximum theoretical suppression available from the third harmonic is defined as the resonance capacity:

$$\text{Resonance Capacity}(n) = \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i} \cos\left(\frac{\pi}{p_i}\right)$$

If the resonance capacity could exceed or match the main density term, a 1-dimensional sieve could theoretically suppress the penalized mass S_2 below S_1. However, the geometry of the Krafft cosine strictly forbids this.

Lemma 4.3 (The Harmonic Deficit [ ]). *For any valid prime window $\mathcal{P}_n$, the resonance capacity of any purely 1-dimensional 3rd harmonic weight system is strictly less than the sieve's main density term:*

$$\sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i} \cos\left(\frac{\pi}{p_i}\right) < \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i} \tag{4.2}$$

Proof. For any prime $p_i \geq 5$, the phase angle satisfies $0 < \pi/p_i < \pi/2$. Within this open interval, the cosine function is strictly bounded by 1. Consequently, the strict inequality $\cos(\pi/p_i) < 1$ holds for every individual term. Multiplying by the strictly positive amplitude $2/p_i$ yields $\frac{2}{p_i}\cos(\pi/p_i) < \frac{2}{p_i}$. Summing these strict, term-by-term inequalities across the finite non-empty set $\mathcal{P}_n$ rigorously preserves the strict inequality for the aggregate sums. $\square$

While the strict inequality itself is algebraically elementary, the significance lies not just in the deficit, but in the spectral signature of the divergence.

Proposition 4.4 (1-Dimensional Parity Collision). *For any sufficiently large n, utilizing purely 1-dimensional independent harmonic waves forces the continuous sieve density quotient to $\mu(n, W) \geq 1$. It is analytically impossible to resolve the Twin Prime Configuration using strictly 1-dimensional variance bounds.*

Proof. *Note: This proof relies on classical continuous analysis (Mertens' theorems) and limit behaviors as $n \to \infty$, which fall outside the scope of the project's discrete Lean formalization.*

By Mertens' theorems [11], the main density term $\sum \frac{2}{p_i}$ diverges logarithmically as $n \to \infty$. Conversely, the magnitude of the harmonic deficit (4.2), measured by the difference between the main term and the resonance capacity, can be bounded using the Taylor expansion $\cos(x) \geq 1 - x^2/2$:

$$\text{Deficit} = \sum_{p_i \in \mathcal{P}_n} \frac{2}{p_i}\left(1 - \cos\frac{\pi}{p_i}\right) \leq \sum_{p_i \in \mathcal{P}_n} \frac{\pi^2}{p_i^3}$$

Because the sum $\sum p^{-3}$ strictly converges, the absolute suppression capacity of the 1-dimensional harmonics falls infinitely behind the diverging main density.

This convergence/divergence disparity forces the quotient $\mu \geq 1$ for any sufficiently large n when utilizing 1-dimensional independent waves. A finite truncation of purely independent harmonics will force the weight function to take negative values at dense overlapping forbidden regions before it can reduce the overall Rayleigh quotient below 1. □

This establishes a barrier. To circumvent this continuous density deficit, the weight geometry must inherently incorporate multidimensional cross-harmonic interactions.

5. Topological Reduction to the Minimum Spectral Ratio

The analytic collision formalized in Section 4 dictates that utilizing purely independent 1-dimensional harmonic waves fundamentally fails to bypass the continuous density deficit. To constructively overcome this, we must transition to a correlated, multidimensional weight space. We build this space using polynomials of the harmonic base variables.

Definition 5.1 (Multidimensional Basis and Polynomial [[illegible] & [illegible]]). For any subset of prime indices $S \subseteq \{1, \ldots, w(n)\}$, we define the multidimensional basis function as the product of the third-harmonic cosines:

$$B_S(x) = \prod_{i \in S} \cos\left(\frac{6\pi x}{p_i}\right)$$

Let λ be a real-valued coefficient vector indexed by the power set of $\{1, \ldots, w(n)\}$. We construct the multidimensional polynomial $P(x)$ as a linear combination of these basis functions:

$$P(x) = \sum_S \lambda_S B_S(x)$$

The resulting $2^{w(n)}$-dimensional basis is closed under pointwise multiplication, which allows us to enforce the strict non-negativity constraint of the Krafft sieve.

Definition 5.2 (The Continuous Weight Function [[illegible]]). We define the multidimensional weight function $W_\lambda(x)$ by squaring the polynomial $P(x)$ and strictly restricting its support to the evaluation interval $\mathcal{A}_n$:

$$W_\lambda(x) = \begin{cases} (P(x))^2 & \text{if } x \in \mathcal{A}_n \\ 0 & \text{otherwise} \end{cases}$$

By casting the weight space as the square of a polynomial, we automatically satisfy the condition $W_\lambda(x) \geq 0$ for all x, circumventing the need for an intractable continuum of linear inequality constraints on the parameter vector λ. This formulation allows us to translate our spatial metrics entirely into the language of linear algebra.

Lemma 5.3 (Equivalence of Moments and Quadratic Forms [? & ?]). *The spatial mass $S_1(n, W_\lambda)$ and the penalized mass $S_2(n, W_\lambda)$ map directly to continuous quadratic forms $Q_1(\lambda)$ and $Q_2(\lambda)$. Specifically, the total mass evaluates to the squared norm of the polynomial over the interval:*

$$S_1(n, W_\lambda) = Q_1(\lambda) = \sum_{x \in \mathcal{A}_n} (P(x))^2$$

Proof. By definition, the total spatial mass is $S_1 = \sum_{x \in \mathcal{A}_n} W_\lambda(x)$. Substituting the construction $W_\lambda(x) = P(x)^2$ expands algebraically to $\sum_x (\sum_S \lambda_S B_S(x))^2$. Distributing the product of the finite sums and rearranging the summation order by Fubini's theorem isolates the cross-terms of the basis functions, defining the matrix elements $M_1(S, T) = \sum_x B_S(x) B_T(x)$. This yields the explicit quadratic form $Q_1(\lambda) = \sum_{S,T} \lambda_S M_1(S, T) \lambda_T$. The derivation for $S_2(n, W_\lambda) = Q_2(\lambda)$ follows identically by incorporating the linear hit counter $c(x)$ into the inner sum. $\square$

With our metrics translated into quadratic forms, the sieve density quotient (4.1) transitions to the continuous Rayleigh quotient $R(\lambda) = Q_2(\lambda)/Q_1(\lambda)$ [?]. Because this quotient is inherently scale-invariant, we reduce to a minimisation problem on a compact set.

Lemma 5.4 (Compactness of the Orthogonal Unit Sphere [?]). *Let* $\ker(Q_1)$ *denote the kernel of the quadratic form Q_1. Because the Rayleigh quotient is invariant under both uniform scaling and the additive translation of vectors from* $\ker(Q_1)$*, we restrict the parameter space strictly to the unit sphere within the orthogonal complement, denoted as $\mathcal{S}^\perp$. This bounded subspace $\mathcal{S}^\perp$ is topologically compact.*

Proof. Within the finite-dimensional Euclidean space of the coefficients, compactness is equivalent to the subspace being both closed and bounded. The orthogonal complement $\ker(Q_1)^\perp$ forms a closed linear subspace. The unit sphere condition, enforced by the standard dot product $\lambda \cdot \lambda = 1$, defines a set that is both closed and bounded. Because $\mathcal{S}^\perp$ is the strict intersection of these closed constraints, it inherits the topological properties and is definitively compact. $\square$

Having established Lemma 5.4, we can now rigorously analyze the continuous image of this subspace under the Rayleigh quotient mapping. This allows us to formally define the absolute lower bound of the sieve density quotient (4.1).

Definition 5.5 (The Minimum Attainable Ratio). We define the set of attainable ratios as the continuous image of the compact space $\mathcal{S}^\perp$ under the mapping

$R(\lambda)$. The minimum spectral ratio $\mu_{\min}(n)$ is defined as the absolute infimum of this attainable set.

While an infimum defines a greatest lower bound, it does not inherently guarantee that this bound is attained. However, by leveraging the topological compactness derived above, we can assert the existence of an exact optimal coefficient vector.

Theorem 5.6 (Existence of the Exact Minimizer [ ]). *The minimum spectral ratio $\mu_{\min}(n)$ is strictly attained by an optimal coefficient vector $\lambda_{opt} \in \mathcal{S}^{\perp}$, such that $Q_1(\lambda_{opt}) > 0$ and $R(\lambda_{opt}) = \mu_{\min}(n)$.*

Proof. The Rayleigh quotient $R(\lambda)$ acts as a continuous rational function at all points where its denominator is strictly positive, $Q_1(\lambda) > 0$. By confining our domain strictly to the orthogonal complement $\mathcal{S}^{\perp}$, we universally exclude the kernel of Q_1, ensuring continuity across the entire restricted domain. Because continuous functions map compact topological spaces strictly to compact spaces, the resulting set of attainable ratios is itself compact. By the Extreme Value Theorem, a compact subset of the real numbers contains its infimum, guaranteeing the existence of the minimizer λ_{opt}. $\square$

With the existence of the minimiser established, we reach the culmination of the Krafft geometry. We can now formally link the continuous spectral bound $\mu_{min}(n)$ back to the discrete existence of twin primes within our evaluation interval.

Theorem 5.7 (Optimal Admissibility and Sieve Guarantee [&]). *The optimal multidimensional weight function $W_{opt}(x)$ synthesized from λ_{opt} satisfies the strict Definition 3.3 condition ($S_2 < S_1$) if and only if $\mu_{\min}(n) < 1$. If this analytic bound holds, the geometric framework rigorously guarantees the existence of a Twin Prime pair index within $\mathcal{A}_n$.*

Proof. By the definition of the minimum ratio, evaluating the moments with the optimal weight function yields exactly $S_2(n, W_{opt})/S_1(n, W_{opt}) = R(\lambda_{opt}) = \mu_{\min}(n)$. Since S_1 is strictly positive for the optimal non-zero weight, it follows algebraically that $S_2 < S_1$ holds if and only if the ratio $\mu_{\min}(n) < 1$. If this strict inequality is satisfied, the weight function W_{opt} inherently fulfills the Definition 3.3 condition. By directly applying the previously established Theorem 2.7, this state of structural admissibility forces the existence of at least one index $x \in \mathcal{A}_n$ that strictly generates a twin prime pair. $\square$

This theorem completes the geometric reduction. The parity obstruction has been translated into an analytic condition: evaluating the spectral density of the cross-harmonic convolutions to confirm $\mu_{\min}(n) < 1$.

6. Discussion: Cross-Harmonic Convolutions and Future Directions

While the reduction of the Twin Prime Conjecture to the spectral bound $\mu_{\min}(n) < 1$ provides a rigid finite-dimensional framework, achieving this bound analytically remains notoriously obstructed by the converging harmonic deficit. However, the multidimensional polynomial space $\mathcal{S}^{\perp}$ theoretically bypasses this limitation. We can map the precise analytic mechanism of this bypass by examining the full inclusion-exclusion convolution.

6.1. *The Perfect Sieve Weight and Spectral Convolution*

To understand how cross-harmonics circumvent the parity barrier, consider the "perfect" spatial weight function $\omega(x)$, which evaluates exactly to 1 when $c(x) = 0$ and 0 elsewhere. It is given by the pointwise product of the allowed residue indicators:

$$\omega(x) = \prod_{i=1}^{w(n)} \big(1 - g_i(x)\big)$$

Transforming this exact spatial equivalence into the frequency domain over $\mathbb{Z}/q\mathbb{Z}$ produces a direct spectral blueprint. Let $\hat{g}_i(h) = \mathcal{F}\{g_i(x)\}$ be the Discrete Fourier Transform of the individual hit sequences. By the Convolution Theorem, the sequence of allowed residues transforms as $\mathcal{F}\{1 - g_i(x)\} = q\delta_{h,0} - \hat{g}_i(h)$. Consequently, the perfect multidimensional weight function in the frequency domain resolves into the full cross-convolution:

$$\Omega_h = \big(q\delta_{h,0} - \hat{g}_1(h)\big) \circledast \big(q\delta_{h,0} - \hat{g}_2(h)\big) \circledast \cdots \circledast \big(q\delta_{h,0} - \hat{g}_{w(n)}(h)\big)$$

where $\circledast$ denotes the circular discrete convolution over $\mathbb{Z}/q\mathbb{Z}$.

When this convolution is expanded, it naturally generates cross-harmonics (e.g., $\hat{g}_i(h_1) \circledast \hat{g}_j(h_2)$). These cross-convolutions are structurally indispensable. Where isolated 1-dimensional waves fail by allowing negative functional overlaps, the cross-correlations act as structurally precise corrective phases. They constructively interfere precisely at the intersections of forbidden congruences, clamping the continuous weight function (as defined in Definition 5.2) tightly at zero. This multidimensional scaling simultaneously preserves non-negativity and protects the main DC density term from over-subtraction.

6.2. *An Analytical Blueprint for the Minimum Ratio*

The exact resolution of this convolution computationally requires navigating the massive $2^{w(n)}$-dimensional polynomial space, making direct computation infeasible for large n. However, establishing Theorem 3.4 unconditionally only requires demonstrating that $\mu_{\min}(n) < 1$. By the continuous Variational Principle, we are not required to locate the exact optimal minimizer λ_{opt}. Any analytically synthesized test vector λ_{test} corresponding to a weight matrix that pushes the Rayleigh quotient strictly below 1 is sufficient.

Advancing the resolution of the Twin Prime Conjecture within this Krafft geometry therefore requires executing the following analytical roadmap:

(1) Expand the theoretical multidimensional convolution Ω_h and extract the lowest-order, highest-magnitude cross-harmonic coefficients.
(2) Map these theoretical phase corrections onto the finite-dimensional polynomial basis $B_S(x) = \prod_{i \in S} \cos(6\pi x/p_i)$.
(3) Synthesize a continuous bounded weight vector λ_{test} over the target interval $\mathcal{A}_n$.
(4) Analytically bound the quadratic limits $Q_2(\lambda_{test})/Q_1(\lambda_{test})$, proving that the scaled cross-harmonics inherently possess the capacity to overcome the converging harmonic deficit.

While executing this final step demands sophisticated analytic machinery capable of resolving extensive nested combinatorial sums, the framework nevertheless provides a well-defined approach. Importantly, this continuous analytic framework definitively maps a combinatorial obstacle into an explicit, resolvable linear algebra problem.

7. Conclusion

This paper has presented a rigorous geometric reduction that equates the existence of twin primes to the bounded eigenvalue optimization of the Krafft residue structure. By tracking localized sieve hits over symmetric bounded intervals and translating them into the frequency domain, we successfully mapped discrete prime alignments to continuous variance spectra.

This continuous formulation allowed us to isolate the exact spectral manifestation of the parity barrier. We proved that 1-dimensional harmonic weights suffer from a continuous density deficit, forcing the sieve quotient $\mu \geq 1$ and structurally necessitating the use of multidimensional polynomial bases. Combining topological compactness with harmonic analysis on geometric intervals, our reduction to $\mu_{\min}(n) < 1$ removes the combinatorial ambiguity of parity boundaries. It refines the twin prime problem entirely into estimating the spectral density of heavily structured cross-harmonic convolutions, providing a concrete framework for addressing the parity barrier.

Acknowledgments

The author acknowledges the use of generative AI (Gemini 3.1 Pro [7]) for LaTeX technical synthesis and structural formatting, as well as for blueprinting the Lean 4 formalization by designing precise prompt specifications for implementation by the

Aristotle [1] automated theorem prover. The core mathematical logic, proofs, and final validation were performed by the author.

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